

bioassist.ai

Platform Blueprint

A clean, user-controlled health operating system that aggregates data, quantifies trajectories, and produces evidence-linked recommendations.

Design goals

- Start with near-zero friction; expand to deep clinical detail only when the user wants it.
- Unify messy health data into a single, queryable timeline with provenance.
- Generate recommendations that are **actionable**, **measurable**, and **evidence-linked** (with citations and uncertainty).
- Operate as an assistant, not a clinician: safety checks, escalation, and clear limitations.

Version: 2026-03-05 | Audience: product + engineering + clinical advisors

Educational and product-design use only. Not medical advice.

1. Executive summary

BioAssist is a personal health platform that ingests health information from documents, devices, and connected records, normalizes it into a canonical health timeline, and produces prioritized recommendations with measurement plans.

The core loop

- **Collect** (documents, labs, wearables, visits, self-reports).
- **Normalize** (units, reference ranges, codes; resolve duplicates; attach provenance).
- **Understand** (trends, variability, risk scores, symptom clusters, context).
- **Recommend** (actions + monitoring) with a clear evidence grade and expected outcome impact.
- **Re-measure** to close the loop and learn.

This blueprint maps the content of the playbook PDFs into a product architecture: biopsychosocial intake, measurement checklist, evidence map, test/device atlas, and a longitudinal workflow.

2. Product principles

Progressive disclosure	The UI should show only what is needed now. Advanced panels and tests appear when relevant.
User ownership	Export, delete, and share controls are first-class. Every data point has a source.
Evidence transparency	Recommendations are paired with citations, evidence grade, and expected effect size range.
Low-burden first	Prefer low-risk, high-signal measurements before invasive or expensive diagnostics.
Safety guardrails	Red-flag triage, contraindication checks, and clinician escalation built in.
Longitudinal by default	Trend lines, variability, and adherence beats one-time snapshots.

3. Users and primary use cases

Design for a spectrum: from a person who only wants a clean dashboard to a technical user who wants deep analytics and research citations.

Personas

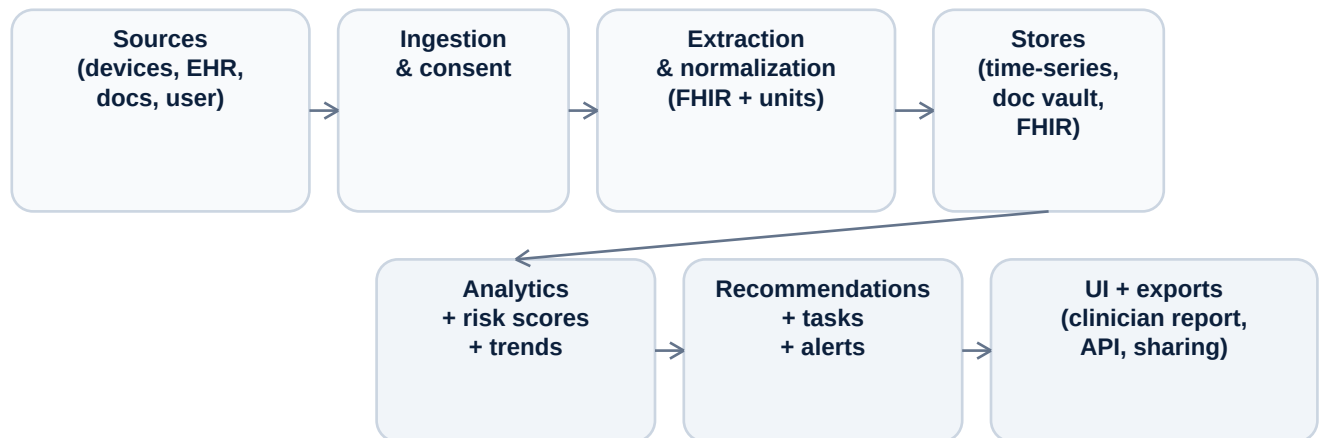
- **Minimalist tracker:** wants a small weekly dashboard (sleep, activity, weight, BP) and a short list of next actions.
- **Chronic-condition manager:** wants labs, meds, symptoms, and device data (e.g., CGM, BP) with alerts and care-plan tracking.
- **Performance / longevity optimizer:** wants testing strategy, fitness metrics, experiment mode, and evidence maps.
- **Care coordinator:** manages records for a child/parent and needs sharing, appointments, and document organization.

Top jobs-to-be-done

- Upload a stack of PDFs and instantly get a structured timeline: labs, imaging, diagnoses, meds, procedures.
- Answer questions like: “What happened around the time my fatigue started?” or “How has LDL changed since 2022?”
- Get a prioritized “next best actions” list: screenings, follow-up labs, behavior levers, clinician discussions.
- Maintain a single source of truth for meds, allergies, conditions, and key vitals for new clinicians.
- Run N-of-1 experiments (diet/sleep/training changes) with causal-inference-friendly tracking.

4. System architecture

A health platform must be built around ingestion, normalization, provenance, and safety. The AI layer should sit on top of a well-defined data model, not replace it.



Key idea: LLMs help with extraction, summarization, and conversational UX, but the source of truth is structured data with provenance and deterministic computation.

5. Data model and normalization

Use a canonical internal model aligned to HL7 FHIR (resources like Patient, Observation, Condition, MedicationRequest, DiagnosticReport, Procedure, ImagingStudy, DocumentReference). Store raw artifacts and extracted structure side-by-side.

Stores

- **Document Vault:** original uploads (PDFs/images), immutable, versioned, encrypted.
- **FHIR-like Clinical Store:** normalized events (labs, diagnoses, meds, encounters) with codes (LOINC, SNOMED, RxNorm where possible).
- **Time-series Store:** high-frequency device data (steps, HR, CGM, sleep stages, SpO2).
- **Knowledge Store:** guidelines, evidence summaries, citations, and your own playbook content.
- **Feature Store:** derived metrics (trends, variability, risk scores, adherence).

Normalization responsibilities

- Unit harmonization (e.g., mg/dL vs mmol/L) and reference range capture (lab-specific).
- De-duplication across sources (same lab result in PDF and EHR feed).
- Time alignment and timezone normalization (including “unknown time” handling).
- Provenance graph: every derived metric links back to raw sources and extraction steps.
- Confidence scoring: extraction confidence, source trust, and recency.

6. Ingestion and integrations

BioAssist should support three ingestion modes from day one: (1) connect a device platform, (2) upload documents, (3) manual quick-entry. Users can stop at any level.

Integrations (prioritized)

- **Mobile health aggregators:** Apple HealthKit / Health Records, Android Health Connect.
- **Wearables:** Oura, Whoop, Fitbit, Garmin, Withings, Apple Watch (via HealthKit).
- **Continuous glucose:** Dexcom/Abbott where user-authorized APIs are available; otherwise import exports.
- **Labs:** PDF ingestion first; later direct connections where available (patient portals vary).
- **EHR:** SMART on FHIR connections when supported by provider portals; Blue Button/claims where applicable.
- **Receipts:** email forwarding + photo upload (pharmacy, supplements, fitness tests).

Document ingestion pipeline

- Classify document type (lab report, imaging report, visit note, discharge summary, receipt).
- Extract structured fields using a hybrid approach: OCR + rules + LLM structured output.
- Map tests to LOINC when possible; preserve original labels when not.
- Detect abnormal flags, capture reference ranges, and attach lab name and specimen type.
- Human-in-the-loop correction UI: quick review for low-confidence fields.

7. Information architecture and pages

The UI should feel like a health “home screen”: a small set of primary tabs, with deep drill-down available but not required.

Home	Today card, key vitals, current priorities, recent changes, red flags.
Timeline	Unified chronological feed of events: labs, visits, symptoms, meds, device summaries.
Insights	Trends, correlations, variability, risk scores, and domain summaries (CV, metabolic, sleep, mental).
Plan	Prioritized recommendations, tasks, goals, measurement schedule, experiment plans.
Documents	Vault of uploads, parsed summaries, provenance, export/share controls.
Devices	Connected sources, data quality, permissions, sync health, per-device views.
Learn	Evidence map, guideline snippets, citations, “why this matters” explanations.
Settings & Privacy	Data permissions, sharing, deletion, export, audit log, clinician mode.

Mapping to the playbook: “Home/Insights/Plan” correspond to the playbook’s minimal dashboard, domain modules, and intervention workflow. “Learn” surfaces the evidence map and device/test atlas.

8. Page blueprints (what is on each page)

Each page is a set of reusable Material-style components: cards, charts, and action lists. Avoid complex navigation; prioritize search + a single global timeline.

8.1 Home

Primary components

- **Priority stack:** 3-7 cards (screenings due, abnormal trend, missing data, adherence opportunity).
- **Weekly snapshot:** sleep, activity, weight/body comp, BP, glucose (if available), mood/stress.
- **Red flags:** only if present; otherwise hidden. Clear escalation language.
- **Data quality bar:** “Coverage: 62% this week” + top missing signals.

Interaction: tap any card to drill down; swipe between weeks; global search.

8.2 Timeline

- A single chronological feed with filters: *labs, imaging, meds, symptoms, encounters, devices*.
- Every item has: date/time, summary, key numbers, source badge (PDF/EHR/device), confidence.
- Fast compare mode: pick two dates, see what changed.
- Export: generate a clinician-facing “last 12 months summary” PDF.

8.3 Insights

- **Domain tabs:** cardiovascular, metabolic, sleep, mental, kidney, liver/GI, neuro, musculoskeletal.
- **Trends and variability:** show trajectories and confidence intervals; flag sustained changes.
- **Risk scores:** computed deterministically; show inputs and sensitivity (what drives the score).
- **Correlation explorer:** user-selectable; caution language (correlation != causation).
- **Explainability:** “What does this mean?” cards with citations.

8.4 Plan

The Plan is the product's heart: it turns data into actions. Actions must be measurable and time-bound, and they must say *why* and *how to verify*.

Action types

- **Behavior levers:** sleep schedule, activity targets, nutrition changes, stress practices.
- **Clinical follow-ups:** screening due, repeat test, medication review prompts.
- **Measurement adds:** “Add BP cuff weekly” or “Consider HSAT screening” when risk signals appear.
- **Experiments:** structured N-of-1 protocols (baseline -> intervention -> washout).

Each recommendation card includes: expected outcome affected, evidence grade, estimated effect size, risks/downsides, cost range, and a measurement plan.

8.5 Documents

- Folderless by default: use tags and search (lab, imaging, visit note, receipt).
- Auto-generated 1-page summary for each document + extracted structured fields.
- Side-by-side view: original PDF on left, extracted table on right, edit controls.
- Provenance: show exactly which page/region a value came from.
- Sharing: generate expiring links or a clinician packet export.

8.6 Devices

- Connection center with last-sync status, permissions, and data coverage metrics.
- Per-device data contract: what it provides, sampling rate, known limitations (e.g., wrist PPG).
- Calibration and validation prompts (e.g., BP cuff accuracy check).
- Import/export: allow CSV/JSON uploads for devices without APIs.

8.7 Learn

- Evidence map browser: interventions -> outcomes -> effect sizes -> citations.
- Device/test atlas browser: measurement options ordered by invasiveness, cost, and signal quality.
- Guideline snapshots: preventive screening, immunizations, high-value care principles.
- Glossary: biomarkers, imaging, and risk scores explained at multiple depths.

8.8 Settings & Privacy

- Permissions: per-integration + per-data-type toggles.
- Export: FHIR bundle export + human PDF export.
- Delete: account deletion and per-document deletion.
- Audit log: what was accessed, by whom, and when (including AI actions).
- Model controls: opt-in/out of cloud AI, local-only mode where possible.

9. Recommendation engine

Recommendations should be produced by a layered system: deterministic rules and calculators for safety-critical logic, plus an AI layer for explanation, personalization, and narrative synthesis.

Layered decision stack

- **Layer 0 - Safety triage:** red-flag thresholds and symptom patterns trigger “seek care” messaging (no ambiguity).
- **Layer 1 - Guidelines and scheduling:** age/sex/risk-based preventive reminders and medication review prompts.
- **Layer 2 - Risk models:** compute validated scores (e.g., ASCVD inputs) and show driver analysis.
- **Layer 3 - Evidence levers:** map user deficits to interventions with outcome effect sizes (from the evidence map).
- **Layer 4 - Personalization:** constraints, preferences, context (time, budget, diet, injuries) shape the plan.
- **Layer 5 - Experimentation:** propose short cycles to test what works for the individual.

Recommendation data contract

- **Trigger:** what data caused this recommendation.
- **Target:** what outcome/metric it aims to change.
- **Expected effect:** plausible range (not a single number) + uncertainty.
- **Evidence:** citations + grade (RCT/meta, observational, mechanistic).
- **Risks:** contraindications, side effects, false-positive cascades.
- **Cost/time:** rough range and burden score.
- **Measurement plan:** what to measure, how often, and stop rules.

10. AI and research integrations

Use LLMs for language tasks and fuzzy extraction, but constrain them with schemas, retrieval, and verification. Treat every model output as a draft until grounded in sources or structured data.

Where LLMs help

- Document parsing: convert messy PDFs into structured events with confidence scores.
- Clinical note summarization: problem list, medication changes, follow-up instructions.
- Conversational querying: “show my A1c trend”, “what changed after I started medication X”.
- Recommendation explanation: plain-language and technical modes, with citations.
- Research synthesis: retrieve and summarize new evidence (with a human-review workflow).

Guardrails

- Structured output schemas (JSON) with strict validation; reject unparseable outputs.
- Retrieval-augmented generation: all medical claims must link to a source (guideline or paper).
- No free-form dosing or diagnosis. Provide educational context and “discuss with clinician” framing.
- Truthfulness UI: show “source coverage” and flag when an answer is extrapolating.
- Evaluation harness: regression tests on extraction accuracy and recommendation safety.

11. Privacy, security, and compliance

Trust is a product feature. Many users will not upload records unless privacy controls are obvious and credible.

- **Data minimization:** collect only what the user authorizes; default to local processing where feasible.
- **Encryption:** in transit (TLS) and at rest; separate keys; rotate and audit.
- **Access control:** least privilege; short-lived tokens; per-integration scopes.
- **Auditability:** user-visible access log and export history.
- **Regulatory posture:** design for HIPAA-grade security if handling PHI with covered-entity partners; clear disclaimers and safe messaging.

Safety UX patterns

- When high-risk patterns are detected, show a short, direct message with escalation options (call clinician, urgent care).
- Avoid alarming language for mild abnormalities; focus on trend and context.
- Always show: “This is not a diagnosis. If you have symptoms or concerns, contact a clinician.”

12. Roadmap (phased build)

Build the platform as a stable data product first. AI features become dramatically safer and more useful once the underlying timeline and provenance are correct.

MVP (8-12 weeks)	• Account + privacy settings, document vault, PDF/image upload. • Lab report extraction (CBC/CMP/lipids/A1c) with review UI. • Home dashboard + timeline + basic trends. • Clinician packet export (PDF). • Foundational safety disclaimers and red-flag routing.
v1 (3-6 months)	• Device integrations (HealthKit/Health Connect) + coverage metrics. • Medication list, allergies, conditions; simple interaction checks. • Recommendation cards with evidence links (your playbook as seed). • Search + conversational Q&A; grounded in user data. • Sharing workflows (family/caregivers) and multi-profile support.
v2 (6-12 months)	• SMART on FHIR connections where available; claims ingestion. • Imaging report extraction + structured problem list synthesis. • Experiment mode + causal inference helpers + habit tracking. • Continuous monitoring alerts (user-tunable) and care-plan tracking. • Research pipeline with periodic literature refresh and human curation.

13. Success metrics

- **Activation:** % of users who connect at least one source or upload one document within 24 hours.
- **Data coverage:** weekly coverage score improvement (more complete longitudinal signals).
- **Time-to-insight:** time from upload to a usable timeline and top 3 recommendations.
- **Recommendation adoption:** completion rate for top actions and measurement follow-through.
- **Trust:** opt-in retention and low complaint rate about inaccuracies; explicit “confidence” feedback.
- **Clinical usefulness:** clinician packet reuse rate and reported time saved.

Appendix: mapping to the playbook PDFs

- **Biopsychosocial intake prompts** -> Onboarding questionnaire + context profile.
- **Measurement checklist** -> Home dashboard defaults + data coverage meter.
- **Extended test menu + device atlas** -> Learn -> Measurement Explorer + Plan -> add-measurement suggestions.
- **Evidence map** -> Learn -> intervention-outcome explorer + citations attached to recommendation cards.
- **Causal inference cheat sheet** -> Experiment mode templates and analysis helpers.